

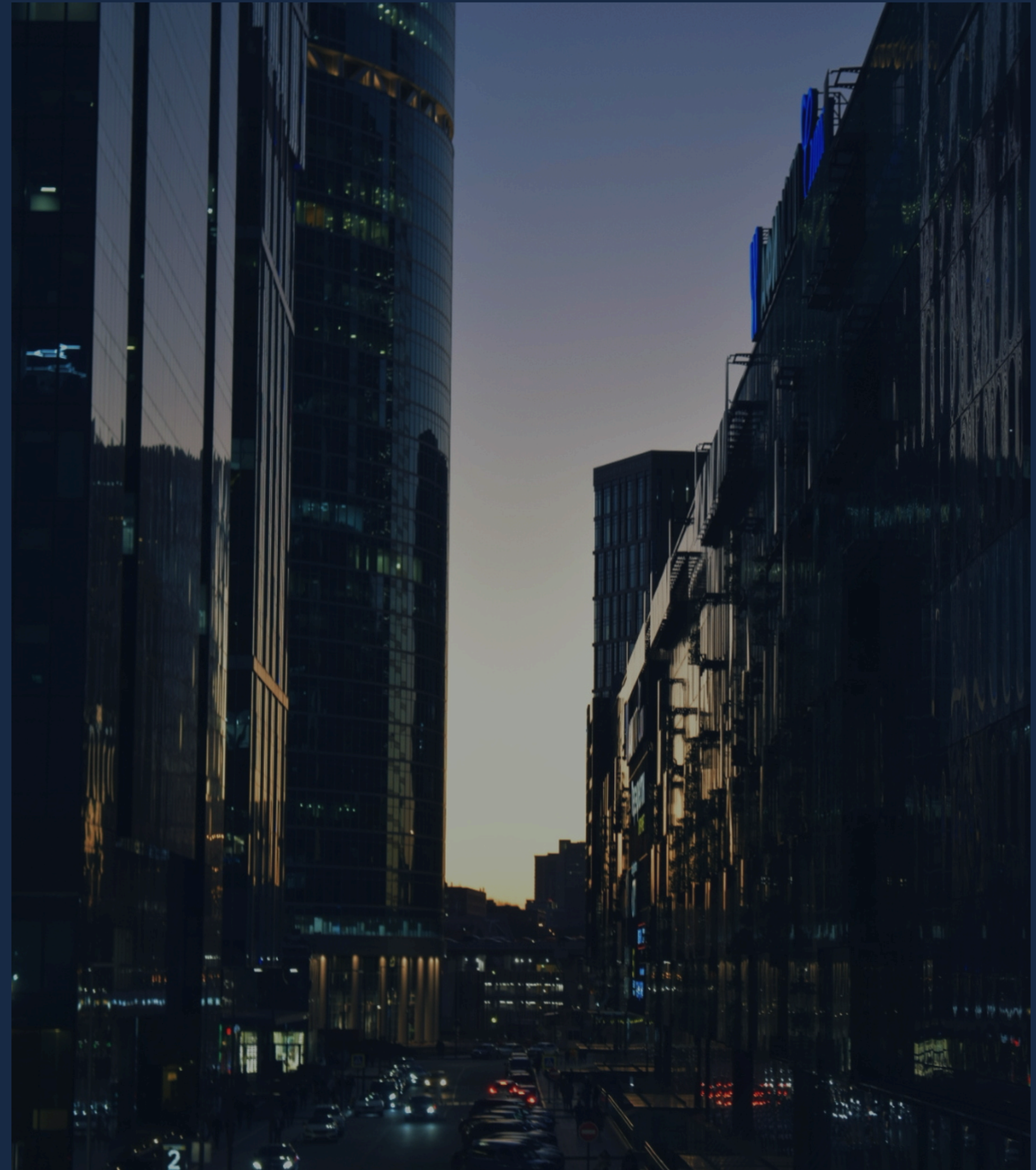
A low-angle, upward-looking photograph of a modern skyscraper at night. The building's glass facade reflects the city lights, and its structural elements create a strong sense of height and scale. The sky is dark, and some windows are illuminated from within.

INVESTMENTS AND PORTFOLIO MANAGEMENT

GROUP 18

AGENDA

1. Investor Profile
2. The Anchoring Bias
3. Signal and Implementation Rules
4. The Efficient Frontier Setup
5. The Optimal Portfolio
6. Theoretical vs Implemented Weights
7. Backtesting Methodology
8. Performance Summary and Discussion
9. StockTrak Implementation
10. Live Performance Evaluation and Risks
11. Brinson Attribution
12. Conclusion



Behavioral Calibration — Who Are We as Investors?

- **BRTI questionnaire: 10 questions, scale 1–7**
 - Completed collectively by the 5-member team
- **Two behavioral parameters derived:**
 - $\lambda = 3.67 \rightarrow$ risk-aversion (below avg. 4 $\rightarrow$ accept more volatility)
 - $q = 0.58 \rightarrow$ perceived risk (above 0.5 $\rightarrow$ risk seen as opportunity)
- **Implication for strategy choice**
 - Dynamic profile justifies a momentum-based long-short strategy
 - Moderate-to-high volatility tolerance compatible with monthly rebalancing

UTILITY FUNCTION

$$U = E[R_p] - (\lambda/2) \times \sigma^2 p$$

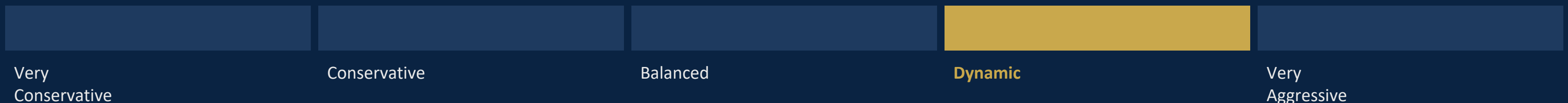
$$\lambda = 3.67 \quad | \quad q = 0.58$$

BRTI SCORE

4.5

DYNAMIC PROFILE

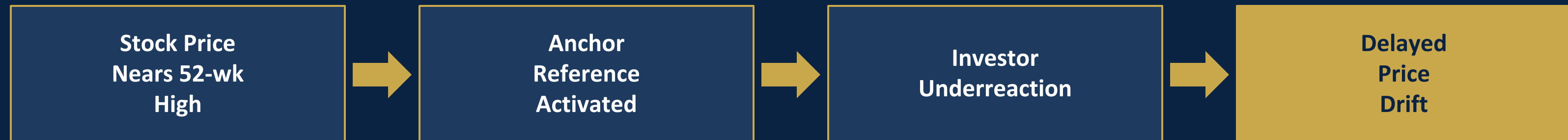
INVESTOR PROFILE SPECTRUM



Why the 52-Week High? The Anchoring Bias

- **Academic foundation:** George & Hwang (2004, Journal of Finance) — proximity to the 52-week high predicts future returns better than standard momentum
- **Behavioral mechanism**
 - Good news fails to push price above the annual high — it feels expensive versus recent memory
 - Result: underreaction → delayed upward drift as information is gradually absorbed
 - Mirror logic on the downside: pessimism persists even after bad news is fully priced in
- **Key insight:** rooted in how humans form judgments under uncertainty — not an arbitrage that vanishes

BEHAVIORAL TRANSMISSION MECHANISM



Signal Design: How We Rank and Select Stocks

SIGNAL FORMULA — for each stock i at month-end t

$$PR(i,t) = P(i,t) / \max(P(i,t-11), \dots, P(i,t)) \quad \rightarrow \quad \text{Ratio} \in (0, 1]$$

UNIVERSE

S&P 500 constituents
Prior year-end snapshot used each January
(avoids look-ahead bias)

FILTERS

\$15 price floor on the short book
Structural asymmetry: short gains capped at 100%,
losses uncapped — early test: -148% in one month
from a regional bank short squeeze

SELECTION & WEIGHTING

Top 10 by PR $\rightarrow$ LONG | Bottom 10 by PR $\rightarrow$ SHORT
Equal weight: 5% per position
Market-neutral by construction

REBALANCING

Monthly at month-end
Continuity tiebreaker: existing positions retained
before substituting new names $\rightarrow$ reduces turnover
and transaction costs

Mean-Variance Analysis :

Built on the 20 stocks (10 longs +10 shorts) from initial portfolio runs from September 2016 to February 2026

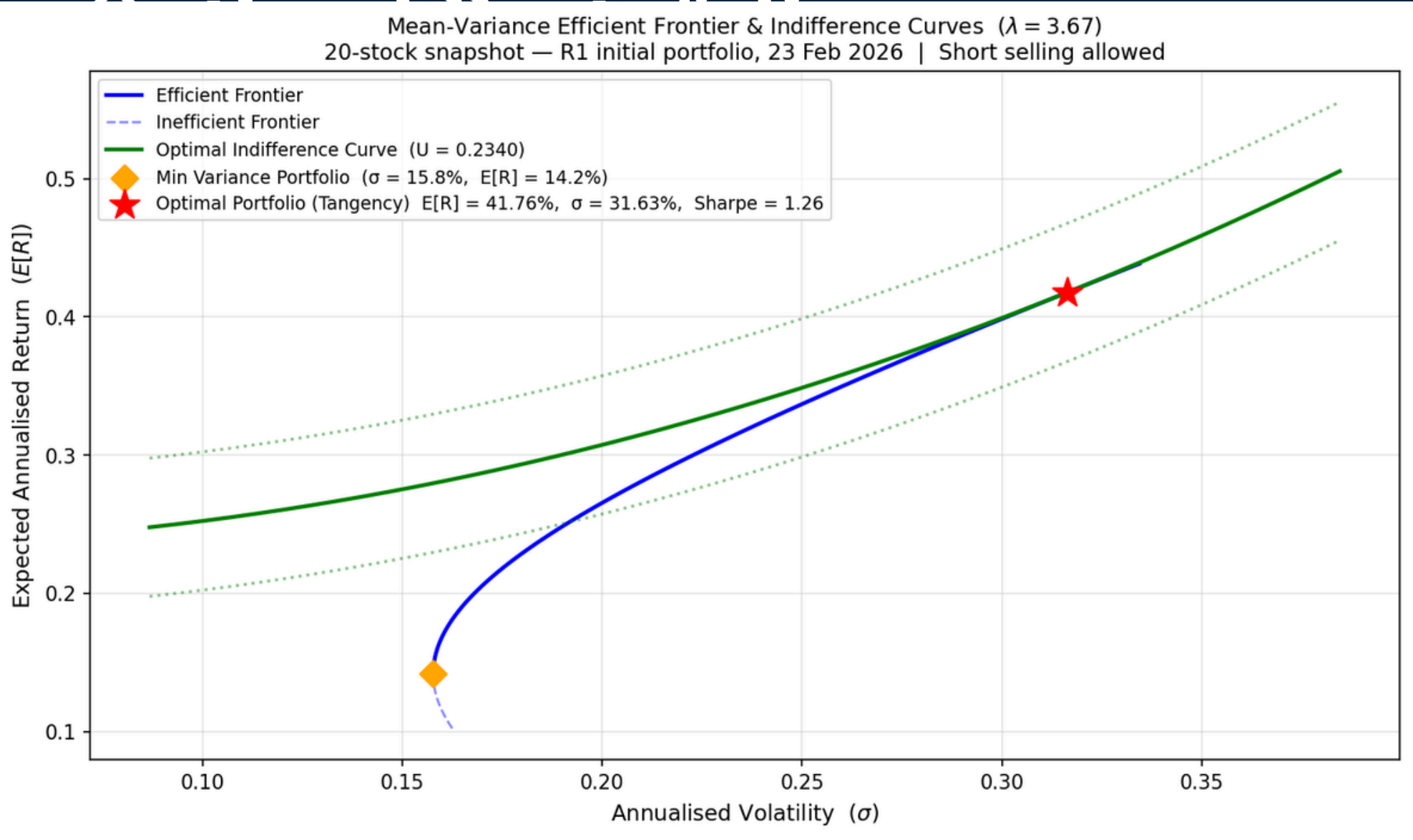
Expected Return : $\mu_i = \bar{r} \times 252$

Risk : $\sum daily \times 252$

Optimization :

Minimize variance across

- target returns
- $w_i \in [-1, 1]$
- Full investment
- $U = E[R_p] - (\lambda/2)\sigma^2 \rightarrow \lambda = 3.67$



EXCEPTED RETURNS :

41.76%

VOLATILTY (σ)

31.63%

SHARPE RATIO :

1.26

OPTIMIMAL INDIFFERENCE CURVE :

0.2340

Theoretical Weights vs. Implemented Strategy

PROBLEM 1 :

Top MVO weights:

Long : KO +39%, KEYS +39%, IRM +36%, GLW +36%, JCI +24%

Short : FDS -51%, KIM -32%

Two positions = 75% of the gross long book.

One position = more than 51% of the short portfolio

→ **Idiosyncratic risk overwhelms diversification**

PROBLEM 2 :

TTD: 52-week high ratio = 0.274 **BUT** strategy signals **short**

Reason : TTD's 10-year historical average return is high

SOLUTION :

- Equal weighting (5%)
- 10 longs + 10 shorts
- Market-neutral portfolio (only returns from signals)

→ **Performance driven purely by the 52-week high signal, not allocation choices**

→ **Eliminates idiosyncratic concentration risk**

→ **Consistent with moderate risk profile ($\lambda = 3.67$)**

Data & Universe

- Jan 2015 – Dec 2025
- 120 monthly observations
- S&P 500 constituents (prior year-end)
- \$15 price filter
- No survivorship bias

Signal & Portfolio

- PR ratio = $\text{Price} \div 52\text{-week max}$
- Long top 10, short bottom 10
- 5% each
- Market-neutral by construction
- Monthly rebalancing

Costs & Split

- \$2/trade on \$1M capital
- Train: Jan 2015–Dec 2022 (84 months)
- Test: Jan 2023–Dec 2025 (36 months)

BACKTESTING METHODOLOGY



PERFORMANCE SUMMARY

Metric	Full Period	Train	Test
Ann. Return	3.36%	2.27%	4.73%
Ann. Volatility	13.95%	15.20%	10.65%
Sharpe (Rf = 2%)	0.097	0.051	0.257
Max Drawdown	-19.94%	-19.94%	-10.32%
Jensen's Alpha	1.70%	0.93%	8.51%
Info. Ratio	-0.410	-0.221	-0.898

DISCUSSION

01 Negative IR is expected

Strategy is market-neutral: zero beta, zero equity premium.

The S&P 500 ran at +21%/year in 2023–2025.

The right benchmark is $R_f = 2\%$, and against that we generate positive alpha.

02 Sharpe improvement train → test

Sharpe rises from 0.051 to 0.257 out-of-sample.

Signal strengthens rather than decays,

consistent with the anchoring hypothesis: behavioral effects persist in high-dispersion markets.

03 Limitations

Signal captures anchoring only, no sector or risk filters.

Monthly rebalancing means stocks can revert before we act.



IS OUTPERFORMANCE STATISTICALLY AND ECONOMICALLY MEANINGFUL?

Statistical significance

- Full backtest period used
- Jensen's alpha t-stat = 0.99
- 95% confidence level: $|t| < 1.96$
- Outperformance not statistically significant
- Possible random fluctuations

Economic meaningfulness

- Arbitrage opportunity already exploited
- Unwanted market exposure
- Transaction costs may wipe out returns

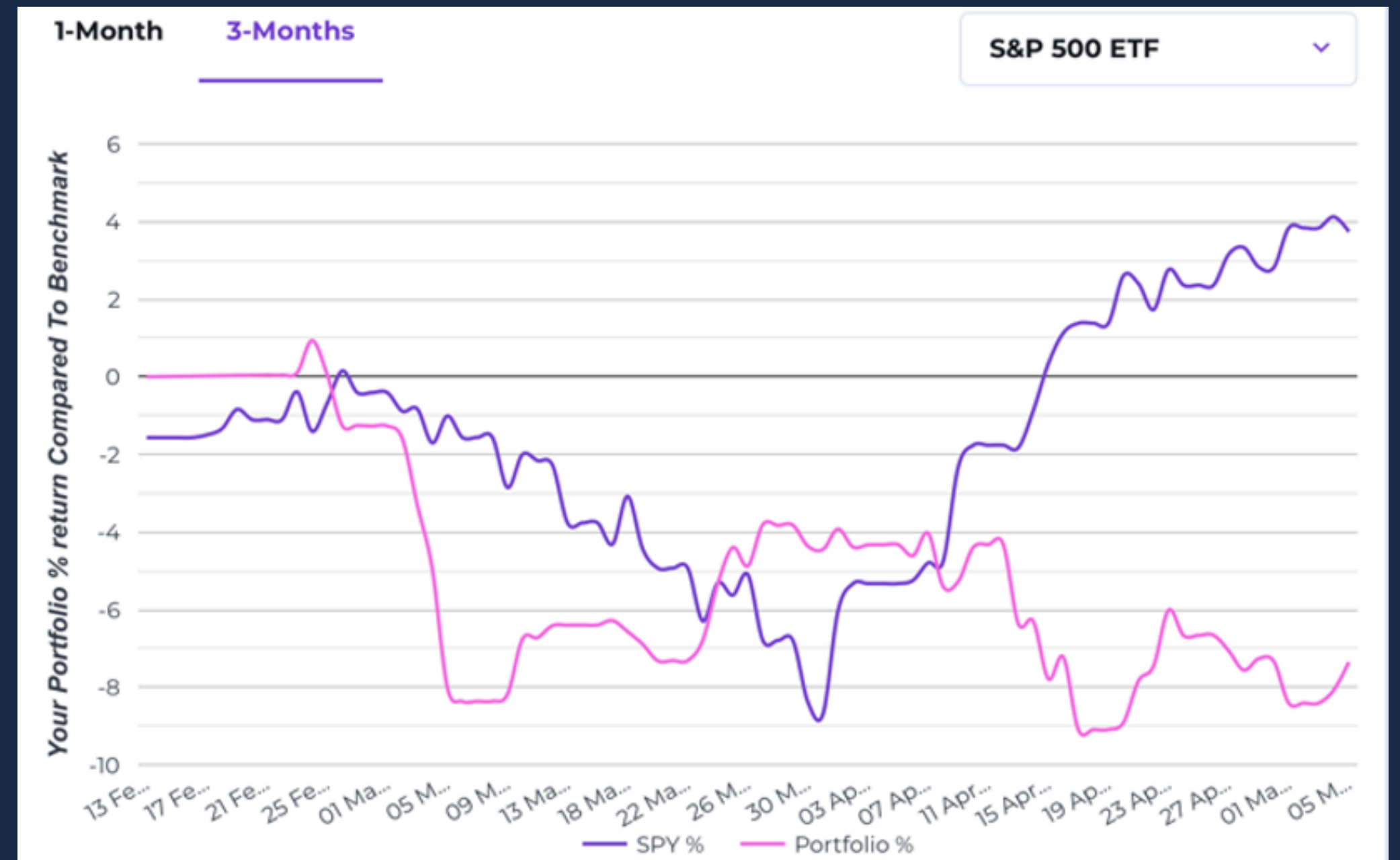
PERFORMANCE : MARKET EXPOSURE (BETA)

Objective: market-neutral portfolio

Final exposure: negative beta

Mouvement: Strong increase
in the S&P 500

Result: decrease in portfolio
performance



PERFORMANCE : SKILL / STRATEGY (ALPHA)

01 Random fluctuation

- Short trading period (3 months)
- High volatility in stock returns

03 Bad implementation :

- Monthly portfolio rebalancing
- Positions adjusted after trend reversals
- Trade-off with higher transaction costs

02 Model Limitations

- Simplistic pricing model
- Exposure to other risk factors
- Momentum factor exposure
- Unstable market conditions

04 Market Efficiency

- Large and efficient market
- Limited mispricing opportunities
- Difficult to exploit pricing inefficiencies
- Better suitability for less efficient markets

DOES THE STRATEGY PERFORM CONSISTENTLY ACROSS SUBPERIODS (TRAIN VS TEST)?

- Positive returns
- Underperformed the S&P 500 Total Return benchmark in both periods
- Excess return:
- Train: -4.21%
- Test: -16.52%

- Positive Jensen's alpha in both periods
- Alpha increased:
- Train: +0.93%
- Test: +8.51%

- Negative market exposure during a strong bull market
- Portfolio beta: -0.29
- Market return: +21.26%

- Lower volatility in test period:
- 10.65% vs 15.20%
- Smaller drawdowns:
- -10.32% vs -19.94%
- Indicates more stable and trending market conditions

DOES THE STRATEGY PERFORM CONSISTENTLY ACROSS SUBPERIODS (TRAIN VS TEST)?

- Market regime risk: strategy performs well in persistent trends but struggles when market conditions change rapidly
- Crash & reversal risk: sudden trend reversals can generate losses on both long and short positions
- Whipsaw risk: in trendless markets, the strategy may buy after price increases and sell after declines
- Crowding risk: widespread adoption of the strategy can amplify mispricing and market corrections
- Short squeeze risk: sharp rebounds in heavily shorted stocks can lead to significant losses
- Market exposure risk: changing or misestimated betas can create unintended directional exposure to the market

Brinson Framework :

	A	B	C	D	E	F	G	H
1	Sector	wP	wB	RP	RB	Allocation	Selection	
2	Communication Services	-5,22%	11,00%	1,86%	-0,15%	0,00024331	-0,0010493	
3	Consumer Discretionary	0,20%	10,00%	-116,77%	-1,10%	0,00107842	-0,0022691	
4	Consumer Staples	11,45%	4,90%	11,85%	14,38%	0,00941784	-0,0028966	
5	Energy	5,13%	3,50%	-3,27%	20,59%	0,00335402	-0,0122377	
6	Financials	5,39%	12,00%	1,34%	-4,42%	0,00292247	0,00310354	
7	Health Care	5,81%	8,50%	-9,59%	1,23%	-0,0003303	-0,006293	
8	Industrials	-10,52%	8,80%	1,08%	11,85%	-0,0228949	0,01133305	
9	Information Technology	3,41%	35,00%	70,72%	-2,28%	0,0072028	0,02488516	
10	Materials	0,00%	1,90%	0,00%	14,92%	-0,0028348	0	
11	Real Estate	-6,16%	1,90%	6,69%	8,05%	-0,006487	0,00083789	
12	Utilities	0,00%	2,30%	0,00%	9,26%	-0,0021298	0	
13	Cash	90,51%	0,00%	0,00%	0,00%	0	0	
14	Total	100,00%	99,80%			-0,010458	0,0154139	0,00495587

ALLOCATION EFFECT :

-1.04%

SELECTION EFFECT :

1.54%

TOTAL EXCESS RETURN :

0.49%

Key drivers:

- IT Selection : +2.49% → JBL +39% vs XLK -2.28%
- Industrials Allocation : -2.29% → short in a sector up +11.85% (XLI)
- Energy Selection : -1.22% → VLO -3.27% vs XLE +20.59%
- Financials : +0.60% → underweight in a sector down -4.42% (XLF)

→ Stock selection added value while Sector allocation was negative
→ The cash position reflects the offset between longs and shorts

CONCLUSION

Key takeaways :

- The 52-week high strategy is based on the anchoring bias
- The strategy generated positive absolute returns but underperformed the S&P 500 benchmark
- Results suggest some evidence of alpha, but not statistically significant
- Negative market exposure reduced return during bull market
- Main risks : regime dependence, reversals and market exposure





THANKS

FOR YOUR ATTENTION